A Mathematical Definition of LLM Outputs

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Abstract

Large Language Models (LLMs) are commonly interpreted as probabilistic next-token generators. However, this interpretation alone does not fully characterize the semantic nature of model outputs, especially under incomplete information. In this work, I propose a mathematical framework that defines LLM outputs as information-conditioned projections over latent semantic states. Instead of treating model outputs as deterministic truths, I formalize generation as an inference process conditioned on an evolving information filtration. I introduce the concept of an Information-Neutral Measure, under which LLM outputs can be interpreted analogously to conditional expectations in financial mathematics and prediction markets. Under this framework, hallucination is not merely a system failure, but a natural consequence of incomplete information prior to oracle observation. This perspective unifies ideas from probabilistic inference, information theory, prediction markets, and generative AI into a common mathematical structure. My formulation further suggests experimentally verifiable hypotheses relating information completeness, retrieval augmentation, and hallucination reduction. I argue that LLMs should fundamentally be viewed not as truth engines, but as information aggregation systems operating over latent semantic states.

1 Introduction

Modern Large Language Models (LLMs) have demonstrated remarkable capabilities across reasoning, coding, retrieval, and generation tasks. Existing interpretations typically model LLMs as autoregressive probabilistic systems:

$$P(y_t \mid y_{<t}, x) \tag{1}$$

where generation is framed as next-token prediction conditioned on prior context.

While operationally effective, this formulation does not fully capture the semantic uncertainty underlying model outputs. In particular, hallucinations reveal that LLM outputs cannot always be interpreted as deterministic representations of reality.

In this paper, I propose a higher-level mathematical interpretation of LLM outputs. I argue that LLM outputs are not deterministic truths, but information-conditioned projections over latent semantic states. I formalize this interpretation using an information-theoretic framework inspired by probability theory, stochastic processes, and prediction market pricing models.

I define the central formulation as:

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) \tag{2}$$

where:

- P_t denotes the generated output at time t,
- Ψ represents the latent semantic truth state,
- $\mathcal{F}_t$ denotes the available information filtration (σ-algebra stream),
- QI is an Information-Neutral Measure,
- Π is an information-conditioned projection operator.

This framework suggests that hallucinations naturally emerge when the information filtration is insufficient to collapse uncertainty regarding the latent semantic state.

2 Information State Space and Core Definitions

I define an information state space by the triplet:

$$(\Omega, \mathcal{F}, Q_I) \tag{3}$$

where Ω is the semantic sample space, $\mathcal{F}$ is the global information σ-algebra, and Q_I is the Information-Neutral Measure.

To bridge prediction markets with generative AI, I formally introduce and adapt the following core definitions mapped directly into LLM-centric terminology:

1. **Information (Context Stream $\mathcal{F}_t$):** The historical sequence of tokens, prompts, and retrieved knowledge blocks available up to step t. It acts as the filtration mapping the model's observable universe.
2. **Information Closure (Contextual Completeness $\mathcal{C}$):** The state where the sigma-algebra $\mathcal{F}_t$ contains sufficient entropic structure to uniquely resolve the true latent state Ψ, meaning no additional external prompt can alter the projection.
3. **Event (Semantic State Transition E):** A specific subset of the sample space Ω corresponding to a coherent semantic meaning or a targeted factual statement generated by the model.
4. **Event Contract (Generation Prompt Target $\mathcal{T}$):** A structured constraint or prompt requirement specifying the desired semantic properties of the output, analogous to a financial payoff structure depending on the realized semantic state.
5. **Result (Output Sequence / Final Token y^*):** The observable realization or factual ground truth that emerges once the model output is verified by an external oracle or finalized in text space.
6. **Oracle (Ground-Truth Verifier / Environment $\mathcal{O}$):** An external system, human evaluator, or environment that provides definitive verification of a generated statement, collapsing the probabilistic state into a deterministic result.
7. **Participant (Attention Sub-Layer / Agent Component $\mathcal{P}_i$):** The internal components of the system (such as individual attention heads or multi-agent nodes) that process distinct segments of the distributed training distribution.

8. **Belief Function (Predictive Belief / Logit Mapping $\mathcal{B}$):** The model's internal subjective probability distribution (represented via softmax logit space) over potential outputs prior to information collapse.

Unlike classical probabilistic language modeling, the semantic state Ψ is not assumed to be directly observable prior to oracle verification. Instead, generation occurs conditionally on partial information.

3 Mathematical Definition of LLM Outputs

I define LLM outputs as projections over latent semantic states conditioned on available information. The core definition is given by:

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) \tag{4}$$

This formulation generalizes traditional autoregressive token probabilities into semantic-state inference. Under this interpretation, components such as prompts, retrieved documents, model parameters, memory systems, tool outputs, and external environments all contribute to the filtration $\mathcal{F}_t$. Thus, generation is fundamentally an information aggregation process driven by the internal *Predictive Belief* $\mathcal{B}$.

4 Hallucination as Incomplete Information Projection

Traditional discussions often treat hallucination as an algorithmic or structural model defect. In contrast, I define hallucination mathematically as the condition:

$$\Pi_{QI}(\Psi \mid \mathcal{F}_t) \neq \Psi^* \tag{5}$$

where Ψ^* denotes the oracle-observed reality ($\mathcal{O}$).

Under this definition, hallucination emerges naturally when the available information is incomplete (lacking *Contextual Completeness*), the latent semantic state remains unresolved, or the projection operator is imperfectly calibrated. This interpretation resembles state collapse phenomena in probabilistic physical systems, where uncertainty persists prior to exact observation by the *Oracle*.

5 Relation to Prediction Markets

The proposed framework shares profound structural similarities with prediction market pricing. Prediction markets are often modeled as:

$$P_t = \mathbb{E}_{QI}[X \mid \mathcal{F}_t] \tag{6}$$

where market prices aggregate distributed information across participants.

Similarly, LLMs aggregate distributed information embedded within training corpora, retrieval systems, prompts, and inference-time interactions. Under this perspective, while markets aggregate beliefs over physical events via human trading, LLMs aggregate beliefs over semantic states using internal *Participants* (attention layers). Both systems operate as information aggregation mechanisms over incomplete knowledge spaces regulated by an implicit *Event Contract*.

6 Experimental Hypotheses

My framework yields experimentally testable predictions. Define the hallucination distance metric:

$$H_t = d\left(\Pi_{QI}(\Psi \mid \mathcal{F}_t), \Psi^*\right) \tag{7}$$

where d denotes the semantic distance from the oracle truth. I hypothesize the following monotonic relationship:

$$H_t \downarrow \quad \text{as} \quad |\mathcal{F}_t| \uparrow \tag{8}$$

That is, hallucination decreases monotonically as information completeness increases toward *Information Closure*. This prediction can be evaluated empirically using retrieval augmentation, tool-augmented inference, external memory systems, multi-agent reasoning, and prediction-market-informed generation.

7 Implications

This framework suggests several broader implications for the field of artificial intelligence:

1. LLMs should not be interpreted as deterministic truth engines.
2. Uncertainty and hallucination are not accidental anomalies, but intrinsic properties of incomplete information systems before reaching *Information Closure*.
3. Future AI systems may benefit from explicitly modeling semantic uncertainty, information completeness, and oracle verification processes.
4. This perspective provides a potential bridge between generative AI, probabilistic inference, information theory, and financial information aggregation systems.

8 Conclusion

I introduced a mathematical framework interpreting LLM outputs as information-conditioned projections over latent semantic states. Rather than viewing generation as deterministic truth production, my framework models LLMs as information aggregation systems operating under incomplete information. This perspective provides a formal interpretation of hallucination, a clear connection between prediction markets and AI inference, and experimentally testable hypotheses relating information completeness to semantic accuracy. I believe this formulation may offer a useful theoretical foundation for future work in generative AI, uncertainty modeling, and information-theoretic machine learning.